

LAB 4_ Assignment: Implementation of AE and VAE with

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Full Marks-100 (50+50)

The following assignment has been designed to help you digest the basic concepts as well as practical implementation of one of the most important generative AI tools- AE and VAE.

Recommended Datasets:

Primary Dataset: Fashion-MNIST

- 60,000 training + 10,000 test images
- 28x28 grayscale images
- Perfect for autoencoders due to structural similarities between items
- More challenging than MNIST but still manageable

❖ Alternative Datasets:

- MNIST: Good for initial testing and validation
- CIFAR-10: More complex, good for testing robustness
- CelebA: Excellent for advanced experiments (face reconstruction)

Hints:

Experimental Setup:

A. Standard Autoencoder:

❖ Architecture:

- Encoder: Conv2D layers (32->64 filters)
- Latent space: 64 dimensions
- Decoder: Mirror of encoder with transposed convolutions
- Loss: Mean Squared Error
- Optimizer: Adam with learning rate 0.001

B. Variational Autoencoder:

❖ Architecture:

- Similar convolutional structure
- Additional layers for mean and variance estimation
- Latent space: 64 dimensions
- Loss: Reconstruction + KL divergence
- Reparameterization trick implemented

Training Parameters:

```
{  
'batch_size': 128,  
'epochs': 50,  
'learning_rate': 0.001,  
'validation_split': 0.2,  
'latent_dimensions': [32, 64], # test both  
}
```

Evaluation Metrics:

- Reconstruction loss
- For VAE: KL divergence
- Visual inspection of reconstructions
- Latent space visualization
- Generation quality (for VAE)

Recommended Experiments:

a) Model Comparison:

```
experiments = [  
    {'model_type': 'AE', 'dataset': 'fashion_mnist', 'latent_dim': 64},  
    {'model_type': 'VAE', 'dataset': 'fashion_mnist', 'latent_dim': 64},  
]
```

b) Latent Space Analysis:

```
latent_dims = [16, 32, 64, 128]
```

c) Architecture Variations:

- Try different number of layers
- Experiment with filter sizes
- Test different activation functions

To run these experiments:

1. Start with Fashion-MNIST
 2. Train both AE and VAE models
 3. Compare reconstruction quality
 4. Analyze latent space representations
 5. Test generative capabilities (VAE)
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